TRANSCRIPTION

Jennon Mangawa :

What was your initial role in the nuclear medicine field or laboratory? Right after graduation and passing the board exam, do you immediately become a nuclear medicine technologist? So, are you currently working in the nuclear medicine facility?

- So, right now, I'm a full-time faculty for the BS nuclear medicine technology. So, indirectly related to a nuclear medicine facility but then I'm still in touch as a nuclear medicine technologist.

So, did you try to work in the nuclear medicine laboratory? Regarding that, what was your role when you worked in the nuclear medicine laboratory? how long have you worked in total?

- Yes, Okay, so, I gave my son a year. So, in 2015, that was my passing. So, right after December, I was hired as a nuclear medicine technologist. So, for more than five years as a nuclear medicine technologist in one facility and then I was hired as a supervisor or assistant manager for the nuclear medicine department for two years, 2021 to 2023. So far, seven years.

I. During your undergraduate education under the METEC curriculum, to what extent were you exposed to nuclear medicine concepts like radiopharmaceuticals, radiation safety, etc.?

- Okay, so, particularly for this question, for me, the extent is under immunology and serology, particularly in radioimmunoassay, but then more on concepts. . So, other than that, based on my experience, that's the extent.

Follow up:
During your undergraduate education, you just mentioned nuclear medicine topics?

- Yes,. I was exposed under serologic testing or serologic testing

Follow up:
Since you mentioned immunotherapy, in your opinion, does the current METEC curriculum prepare graduates well for working in nuclear medicine, including instrumentation and work processes?

- No. At least in my curriculum before, that's around 2012 up to 2015 curriculum.

Since you answered no, as you began working, what knowledge aspects or skills felt missing from the METEC curriculum?

- Okay. So, that's a lot. First, radiation safety and protection. That's one of the things that felt missing for me. Next is, related to that is radiopharmaceuticals. Next, aside from those instrumentations, the machines under nuclear medicine are very different from what we studied or learned under the MT curriculum. And last are your clinical procedures. All nuclear medicine procedures, that's what's missing. So, the things you mentioned, such as radiation safety and protection, radiopharmaceuticals, instrumentation, and clinical procedures.

ii. Do you think that these gaps in knowledge or skills affect your early performance or confidence in the field when you work?

- Yeah, definitely. And I think it took me around two years to gain my confidence to say that I'm competent enough for the profession. It seems like two years is a long time.

iii. From your perspective, what minimum knowledge or skills from the nuclear medicine field should METEC graduates have before entering the field?

- Okay, so, it's aligned with my previous answer. Number one, radiation protection and safety. Number two, I won't repeat it, radiation physics. The foundational concepts and principles are very important. Number three, instrumentation. Four, radiopharmaceuticals. And five, clinical procedures. Those are the four competencies that are needed.

iii.. Since you mentioned that, in connection with that, are there any topics under the METEC curriculum you find most valuable in your current work or in your previous work? So, if yes, what are those?

- Those included in the curriculum. So, definitely, immunology and serology. Those are the most important. But it's very important for all the subjects. Definitely, human anatomy and physiology helped me. Next is biochemistry. That's important. In our time, I don't know if it's in the curriculum now, we still have pharmacokinetics and pharmacodynamics under pharmacology. So, for me, that's important. Next is clinical chemistry. Because particularly in the topics of endocrinology, that's very useful for me. Now, sir, we don't have pharmacology in our curriculum. But we have I.S., I.N.O.V., biochem, and C.C.
- The bad thing about that is that you only have two C.C.s, clinical chemistry. Because in our time, that's three. I think that's basic. CC. 2, that's instrumentation. That's up to, I forgot, elevation. But CC. 2, that's the bulk of the tests, including electrolytes. Then, blood gas analysis. Yes, I think. Then, CC. 3 is purely endocrinology and toxicology. So, if you ask me which one is that, the three under endocrinology is the most useful.

Follow up:
if I may ask, sir, you're also an alumni of SLU?

- Yes, I'm an alumni of SLU. Now, you only have two C.C.s. They already won the endocrinology in the C.C. 2 finals. Okay. Anyway, I think you are just following the CMO for the MC curriculum. So, I don't think that's a problem. But when it comes to, or connected to endocrinology and technology, I think the emphasis is good when it comes to endocrinology

Iv. During your transition into the nuclear medicine field, what competencies in the nuclear medicine field did you find difficult to learn or adjust to?

- Sorry. It's a good thing you asked. I'll add to your previous question. Very punctual, very important.

Follow up:

So, it's under CC., is that right?

- Yes. And CC 2. CC 3

V. Now, during your transition into the nuclear medicine field, what competencies did you find difficult to learn or adjust to? I'll add to your previous question. The five competencies. Yes?

- That's the most important. More importantly, since I started my training, first is instrumentation. Second is the concepts of clinical procedures.

Follow up:
How did you adjust to the demands and expectations of nuclear medicine?

- Blood pressure. Adjustment. That's what I said earlier. This is my own experience. It took me two years to gain my confidence. So, when I say confidence, it's very little to no supervision. That I can let it go. So, that's it. If you said earlier that it will take a long time, so I can quote you with that. My adjustment is a bit long. So, definitely. Because it's not completely aligned with the medical technology profession. It's a different profession. But it's a big adjustment. Yes.

But what specific technical skills or tasks did you find yourself having to learn independently or on the job once you actually started working in nuclear medicine?

- Machine operation. Guidelines for acquisition. And processing parameters. That's what I learned as I worked. As I already worked. Then, on the side, it was backwards for me. So, I was already in skills. Then, during the development of skills, that's when I learned the foundational concepts or principles. For example, radiation physics.

What are the concepts under instrumentation? What are the components of a machine?

- That's what I learned. And then, the pharmaceuticals. I already learned about pharmaceuticals.

Vi. How exactly did that part of your training help you handle your responsibilities as a medical technologist in the nuclear medicine field today? based on what you mentioned earlier, how exactly did that part of your training help you handle your responsibilities as a medtech in the nuclear medicine field?

- So, since that's what I need and those are the competencies that I need to learn, when I learned that, I was able to handle my role as a medtech. The skills or competencies needed for the profession. That's the most important thing.

Follow up:
How might these adjustments have been easier if addressed earlier in your medtech training?But since you mentioned that nuclear medicine is a whole different world in our profession as medtech, are there any adjustments that were easier to address if you trained it as a medtech?

- Yes, definitely the most familiar thing when I joined was the RIA. So, the pipetting, processing, pre-analytical, analytical, post-analytical. That's the most familiar. Other than

that, how to scan, image, process images, different procedures. Those are the adjustments that I need to do. So, based on your 7 years of experience in the field, what specific aspects of nuclear medicine practice should be added or receive stronger emphasis in the medtech curriculum for medtech students? So, the 5 core competencies that should be added to the curriculum. Those are the foundational competencies that need to be added.

Vii. In your perspective, how should certain professional competencies be strengthened or emphasized in the medtech curriculum to address the gaps in knowledge or skills? Be strengthened? How should certain professional competencies be strengthened? So, you mean, under the curriculum?

- So, if you ask me in my perspective, I'm a full-time professor for nuclear medicine technology under graduate. So, if you ask me, under the curriculum, when it comes to subjects or courses, I don't know the term now, subjects or courses. So, if it's included as 5 distinct or separate subjects or courses, and if it's a matter of integration or emphasis, as a subject or courses, it also has its own teaching and learning approach and assessment. So, technically, it's like 5 regular subjects or courses. It also has quizzes and laboratory activities.
- So, if we compare it to immunology and serology, it has a lecture portion, right? Yes, sir. And it also has a lab. So, if the concept or principles are under lecture, then during the lab, we apply it. Then, we will have assessments on both. There are quizzes in the lecture, and there are laboratory activities. Experimental science can be done in the laboratory
- Since I'm a full-time professional, and I'm also a full-time faculty, I also teach professional courses. I also teach MTAP.
- And I also do a review program for certification examination papers. So, there is a TOS certification examination for NMT or Nuclear Medicine Technology. There are five clusters. That's the composition of the five clusters. So, I think that's cluster one.
- Number two is Regeneration Protection. Number three is Instrumentation and Quality QA QC. Number four is Radio Pharmaceuticals and Pharmaceuticals. And then five are Clinical Procedures. So, those are the five clusters. Can you repeat the last post? Clinical Procedures, cluster five.
- So, if you ask me how I will integrate it in the curriculum, I will add the five subjects, at least the five subjects. Because they are the core competencies needed for the profession
- Now, I'm in my third year as a full-time professor. So, if I will include it as part of my career medicine practice, I've been doing this for 10 years.
- It's been 7 years directly in the Professor Medicine Facility, but 3 years as an academic. Yes.

Follow up:
Where do you teach?

- At the De La Salle Medical and Health Sciences Institute, or the DLSMHSI in Cavite It is still the only institution offering the BSLMT program

Ix. Overall, would you recommend the formal incorporation of nuclear medicine topics in the metacurriculum?

- Overall, would I recommend it? Honestly, not totally. I don't know how to give it a chance. Because, if I will base it on my current practice as an academic, I would suggest otherwise to create a separate undergraduate program such as BSLMT
- But, if you would really incorporate that in the MT practice, there's a lot that needs to be done. So, when it comes to MT curriculum, at least the amount that I get will increase

Follow up:

Is there anything else you want to share about your experience that would help us understand the alignment between the metacurriculum and nuclear medicine practice?

- Yes. This is just my personal opinion aside from what I've said. It's good that my background is in MT. Because, honestly, nuclear medicine procedures are mostly heavy on physiology. So, more than the anatomy, more on physiology. We are very heavy on those concepts. Chemistry, if it's just a part of chemistry and biochemistry, our foundation is very heavy on that. So, yes, we can understand what's happening there. That's why I emphasized endocrinology because that's one topic when it comes to physiology or body function. So, it's very important because nuclear medicine procedures are into the physiology of the body. Also, the physiology of diseases, for example, in oncology
- It's very appropriate as a foundation for the nuclear medicine technology profession or practice. Of course, on the technical side, we need to catch up a lot. If it's just manipulating a machine, I don't know if it's the first day of nuclear medicine technology.

How to process it, things like that?

- I've been doing this for a long time. I'm the one who has OC.That's why I said I've been doing this for two years before I became confident. Because I want it to be perfect. No one can monitor me before I say it's okay.

X. You can recommend someone for our research, who else do you think we should speak to in order to understand this topic better?

- Yes. If it's a specific profession, because it's just NLP, Madam, I have someone I can recommend.
- Concurrently, our advisor is Ma'am LG. Actually, she's our research FRIP. As for her, Ma'am Pat Otoa.
- I think she's from the province. Yes, she is. Do you know Ma'am Patricia Otoa? If she's not in BGH, she's still in Sacred Heart as a prayer medicine technologist.
- But previously working, I have a friend, just a close friend, but she's a doctor now. So, if she fits your criteria, I think I can recommend her. My best friend is Jeylon Marawang.
- I can recommend her to be our alumni. But of course, I'll let her know first. Previously, she worked for St. Luke's QC. Hopefully, she'll fit your research criteria. She's still currently working. I think she's been working there for more than 5 years.She's been working there for 5 years. I think she's been working there for 7 years

Follow up:

Before we conclude this interview, do you have any questions for us? Is there anything you would like to add?

- It's okay.

TRANSCRIPT (March 19, 2026; 6:30-7:00 pm)

Sir Alfred Bayan (RMT-10-260313-1727)

Interviewer (Aborje):

- We could take approximately 20-20 mins, depending on how much information you would like to share.
- You may decline to answer any question or stop participating into interview at any time for any reason
- Video and opening of camera will not be required, and you are allowed to join as audio only.
- Lastly, with your permission po, may I ask to record this interview? …
- You have signed the consent form and non-disclosure agreement and are aware of your rights as a participant.
- Your response as a Medtech professional under the nuclear medicine field is highly appreciated, and rest assured that all information will be kept private and confidential.
- So, may we start recording na po, sir?

Sir Alfred Bayan (RMT-10-260313-1727)

- Okay, go ahead

Interviewer (Aborje):

- So first po, for the background information, we would like to ask po “What was your initial role in the nuclear medicine field or laboratory?”

Sir Alfred Bayan (RMT-10-260313-1727)

- Right after graduation and passing the board exam, immediately, nuclear medicine technologist na.

Interviewer (Aborje):

- So, are you currently working po in the nuclear medicine facility po?

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah, ito yung clinarify ko. So right now, I’m a full-time faculty for the BS Nuclear Medicine Technology. So indirectly siya related to a nuclear medicine facility. But then I’m still in touch as a nuclear medicine technologist.

Interviewer (Aborje):

- So, na-try niyo pong mag-work po sa nuclear medicine laboratory po sir?

Sir Alfred Bayan (RMT-10-260313-1727)

- Yes..Oo..Yes

Interviewer (Aborje):

- Okay, so, regarding that po, what is your role noong nag-work po kayo sa nuclear medicine laboratory?

Sir Alfred Bayan (RMT-10-260313-1727)

- Okay, sige. So, ibibigay ko lang yung ano, by year. Sa 2015, yun yung passing ko (passed the board exam). So, right after, around December, na-hire na ako as nuclear medicine technologist. So, for more than 5 years, as nuclear medicine technologist in one facility, and then I was hired as a supervisor or assistant manager sa nuclear medicine department din for 2 years, 2021-2023.

Interviewer (Aborje)

- So, from that time po, parang how long have you’ve worked po sa like total

Sir Alfred Bayan (RMT-10-260313-1727)

- So far, 7 years

Interviewer (Aborje)

- 7 years po…

Sir Alfred Bayan (RMT-10-260313-1727)

- Oo, 7 years

Interviewer (Aborje)

- So, now that we have established that, we can now move on to our main questions po. We'll focus on your experiences as an RMT who graduated from the medtech program and worked before po sa nuclear medicine laboratory. So, we'll be asking you your general experience, preparation, and curricular suggestions related to nuclear medicine practice po.

Sir Alfred Bayan (RMT-10-260313-1727)

- Okay…

Interviewer (Aborje)

- For the 1st question, during your undergraduate education, under the medtech curriculum, to what extent were you exposed to nuclear medicine concepts like radiopharmaceuticals, radiation safety, etc.?

Sir Alfred Bayan (RMT-10-260313-1727)

- Okay. So, particularly for this question, for me ang extent lang is under Immunology and serology, particularly sa radioimmunoassay. But then more on concepts siya. So, yuon. Other than that, based on my experience, yun lang yung extent.

Interviewer (Aborje)

- So bale parang sa undergraduate po, IS lang po na-mention yung nuclear medicine topics.

Sir Alfred Bayan (RMT-10-260313-1727)

- Yes… Oo. Parang pahapyaw siya under serologic testing

Interviewer (Aborje)

- So, nabanggit niyo po yung ImmunoSero, in your opinion, does the current medtech curriculum prepared graduates well, for working in nuclear medicine including instrumentation and work processes?

Sir Alfred Bayan (RMT-10-260313-1727)

- No. Atleast in my curriculum before, that's around 2012 up to 2015 na curriculum.

Interviewer (Aborje)

- Okay po. Since you answered no, as you began working po, what knowledge aspects or skills felt missing from the medtech curriculum?

Sir Alfred Bayan (RMT-10-260313-1727)

- Okay so medyo madami-dami yan. Una, radiation protection….radiation safety and protection, yan yung ano, isa sa mga pinaka-foundation na kulang for me. Next is… related to that is ito, radiopharmaceuticals. Next, aside from those, uh.. ano pa nga ba?... instrumentations, yung mga machines under the nuclear medicine is very different from… kung ano yung napag-aralan natin or yung mga natutunan under the MT curriculum. And last are yung clinical procedures, so lahat ng mga nuclear medicine procedures, yan yung wala.

Interviewer (Aborje)

- So, yung mg ana-mention niyo po, such as yung radiation safety and protection, radiopharmaceuticals, instrumentation tsaka clinical procedures, do you think na these gapas in knowledge or skills affect your early performance or confidence in the field po noong nag-work po kayo?

Sir Alfred Bayan (RMT-10-260313-1727)

- Yeah, definitely. Ang alam ko lang is, parang ang dating is, noong na-hire ako, doon palang ako natututo. And I think it took me around 2 years to gain my confidence to say that I'm competent enough for the profession.

Interviewer (Aborje)

- Parang mahaba-haba po yung 2 years sir

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah yeah, totoo. Hahahah

Interviewer (Aborje)

- So moving on po sa next question, from your perspective, what minimum knowledge or skills from the nuclear medicine field should medtech graduates have before entering the field?

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah, sige so, aligned siya sa answer ko previously. Number 1, radiation protection and safety. Number 2, hindi ko nabanggit, radiation physics, pinakaimportante, foundational concepts

yan and principles. Number 3 instrumentations. Number 4, radiopharmaceuticals. And 5, clinical procedures. So, lima. Yun yung, I think, core competences na kailangan.

Interviewer (Aborje)

- So since nabanggit niyo po yun, uhm, in connection with that po uhm are there any topics under the medtech curriculum you find most valuable in your current work or like sa work po before?

- So, if yes ano po yun?

Sir Alfred Bayan (RMT-10-260313-1727)

- Yung ano, included sa curriculum? So, definitely, immunology and serology. Yun yung pinaka ano…. Pero very important ang.. kung lahat ng subjects… definitely nakatulong sakin yung human anatomy and physiology. Next is biochemistry, importante yun. Meron noong time naming, di ko alam sa curriculum na ngayon, meron pa kaming pharmacokinetics and pharmacodynamics under pharmacology. So, for me importante yun. Next is clinical chemistry. Kasi particularly sa topic ng endocrinology, yun yung very useful for me.

Interviewer (Aborje)

- Bale ngayon po kasi sir, wala na kaming pharmacology sa curriculum po namin. Pero bale we have po IS, anaphy, biochem tsaka CC po.

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah okay.. okay. Kasi… ang balita ko nga rin kasi wala na ang… dalawa nalang ang CC niyo noh? Clinical chemistry. Kasi noong time namin, tatlo yun. So, anon nga yun, nakalimutan ko yung coverage ng CC1. I think that's basic… instrumentation tas up to… I forgot…. Basta yung CC2, ito na yung bulk ng tests including electrolytes, tapos, blood gas analysis. Yes I think. Tas yung CC3 pure endocrinology and toxicology siya. So kung tatanungin moa ko alin dun? Yung 3, under endocrinology yung pinaka- useful.

Interviewer (Aborje)

- So, if I may ask po sir, alumni din po kayo sa SLU po?

Sir Alfred Bayan (RMT-10-260313-1727)

- Yes.. oo.. alumni…alumni ng SLU

Interviewer (Aborje)

- Ngayon po kasi, bale dalawang CC nalang po tas hinalo na po nila yung endocrinology sa CC2 finals

Sir Alfred Bayan (RMT-10-260313-1727)

- Okay..okay.. Ah finals na..
- Anyway, I think you are just following the CMO for the MT curriculum so, I don't think… walang problem when it comes to or connected to nuclear medicine technology. I think, Maganda yung emphasis when it comes to endocrinology

Interviewer (Aborje)

- Yes po
- So, moving on po sa next question sir
- During your transition into the nuclear medicine field, what competences in the nuclear medicine field did you find difficult to learn or adjust to?

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah sorry, buti nalang inano mo…
- idagdag ko naman sa previous question mo, Venipuncture, very important. Inversion ng TC? Tama ba?

Interviewer (Aborje)

- yes po

Sir Alfred Bayan (RMT-10-260313-1727)

- TC and Hema…okay.. oo... ayun.. pa-dagdag nalang sa previous question mo
- "transition into the nuclear medicine field, what competences in the nuclear medicine field did you find difficult to learn or adjust to"….
- So, i-align ko ulit doon sa previous question, yung limang competencies. Yun yung pinaka-importante. Or importantly, since hands-on na agad ang training ko, machine, so instrumentations. Then pangalawa, yung concepts ng clinical procedures.

Interviewer (Aborje)

- So bale, yung lima pong nabanggit niyo lang din po kanina po sa previous questions?

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah yes… oo

Interviewer (Aborje)

- So just a follow-up question lang po, How did you adjust to the demands and expectations of nuclear medicine practice?

Sir Alfred Bayan (RMT-10-260313-1727)

- Madugo-dugong adjustment yan. Ayun, kagaya nga ng sinabi ko kanina, ito, in my own experience to. Basta it took me 2 years to gain my confidence. So pag sinabi kong confidence, very little to no supervision lang. na pwede na akong pabayaan ganon. So, yun kung sinabi pa nila na medyo matagal tagal. I can quote you with that. Medyo mahaba rin yung aking adjustment. So, definitely kasi parang hindi siya completely aligned sa medical technology profession. Kaya… it's a different profession. Kaya malaking adjustment talaga siya.

Interviewer (Aborje)

- Opo

- So, moving up po sir, following up on that lang, question, uhm what's…bale nabanggit mo na yung iba pero, what specific technical skills or tasks did you find yourself having to learn independently or on the job, once you actually start working po sa nuclear medicine.

Sir Alfred Bayan (RMT-10-260313-1727)

- So, machine operation, uhm mga guidelines para sa acquisition, and processing parameters, yung mga yun. Yung natutunan ko na lang as I work, as I already work

- Tapos on the side, uhm, parang backwards ang naging dating sakin noh. So, nasa skills na ako, tapos during development ng skills, doon ko lang inaaral yung mga foundational concepts or principles. Kunyari, radiation physics, yung mga yan. Paano, ano ba mga concepts under instrumentation. Ano ba yung mga components ng machine. Yan yung mga natutunan ko habang…. and then yung pharmaceuticals, or radiopharmaceuticals… doon ko nalang din natutunan

Interviewer (Aborje)

- So, how exactly did that part of your training helped you handle your responsibilities as medical technologist in the nuclear medicine field today po?

Sir Alfred Bayan (RMT-10-260313-1727)

- Sorry? Come again?

Interviewer (Aborje)

- So, bale sa mga nabanggit mopo kanina,how exactly did that part of your training helped you handle your responsibilities as medtech in the nuclear medicine field?

Sir Alfred Bayan (RMT-10-260313-1727)

- Oh so, since yun yung mga kailangan, yun yung mag competencies na kailangan kong matutunan, uhm, noong natutunan ko na siya, nagampanan ko yung role ko as… and yung skills or competencies needed for the profession. Kasi yun yung pinaka importante.

Interviewer (Aborje)

- So, siguro po, uhm, based sa mga nasabi mo po kanina, how might these, yung mga adjustments, have been easier if addressed earlier in your medtech training po

Sir Alfred Bayan (RMT-10-260313-1727)

- Yes. Definitely, oo

Interviewer (Aborje)

- Pero since parang nabanggit niyo po na parang whole different world yung nuclear medicine sa profession po natin as medtech, parang are there any adjustments na mas naging mabilis po na na-address if uhm, na-train mo siya as a medtech.

Sir Alfred Bayan (RMT-10-260313-1727)

- No, definitely. So, kasi ang pinaka-familiar lang basically pag pumasok ka, yung REA. So, yung pipetting, processing ng ano, pre-analytical, analytical, post-analytical. Yun lang familiar. Other than that, paano mag-scan, mag-image, mag-process ng images, paano (yung) iba't ibang procedures, yun yung mga kailangan kong…. Yun yung mga adjustments na kailangan kong gawin.

Interviewer (Aborje)

- So bale, based on your 7 years po of experience sa field, what specific aspects of nuclear medicine practice should be added or receive stronger emphasis on the curriculum po for medtech students?

Sir Alfred Bayan (RMT-10-260313-1727)

- Ayan, so, i-aalign ko ulit siya sa question, ay sa sagot ko kanina, ito yung five… for competencies. Dapat kung idadagdag siya sa curriculum, yan yung mga foundational competencies na kailanangan idagdag

Interviewer (Aborje)

- So, bale parang parati po nating nababanggit po yung five po… yung radiation safety, radiopharmaceuticals…so actually po, nasa last question na po tayo sa main questions, and actually connected lang din po ito. In your perspective, how should certain professional competencies be strengthened or emphasized on the medtech curriculum to adjust the gaps in knowledge or skills po.

Sir Alfred Bayan (RMT-10-260313-1727)

- Strength? Uhm, medyo, hindi ganon ka-specific yung tanong, i-clarify ko lang. “how should certain professional competencies be strengthened”? So, you mean parang under the curriculum?

Interviewer (Aborje)

- Yes po sir. In your perspective po

Sir Alfred Bayan (RMT-10-260313-1727)

- So, kung ano, tatanungin mo ko sa perspective, kasi currently, I’m a full-time professor na for nuclear medicine technology undergraduate. So, kung tatanungin mo ako, ang ano ko dyan is, under the curriculum, isama siya when it comes to… o as a subject… or a subjects pala rather, or courses. ‘Di ko alam kung anong term ngayon… subjects or courses. Isama siya as a distinct or separate subjects or courses. And kung manner of, uhm, integration siya or emphasis, as a subject or course, meron rin siyang sariling teaching or learning approach, and assessment. So, para siyang technically, isang regular or limang regular subjects or courses. May quizzes din siya, may laboratory activities… yung mga ganon… do you get me? (Interviewer: yes sir) So, parang i-compare natin sa ano, uhm, immunology serology. Meron siya dibang lecture portion? (Interviewer: yes po). Tapos meron din siyang lab,so, kung under lecture, yung concept… principles. Tas doon sa lab, i-apply natin. Tapos doon, magkakaroon tayo ng assessments on both. Mayrong quizzes sa lecture, tas may laboratory activities. Experiments, mga ganon sa laboratory. I think ganon ko siya naiintindihan based on this question.

- I-ano ko nalang din, for context, bakit ko ba nababanggit yun and thank you atleast mukhang okay….uh… maganda yung discussion natin when it comes to your questions. As a… ayun… nabanggit ko na, I'm full-time professor and I also… full-time faculty rather..and I also teach professional courses, I also teach, uhm, parang yung … anong tawag na dun?.... uh… MTAP… (Interviewer: yes po, sa internship)… oo, yung remediation subject ng ano dun eh…. I also teach remediation subjects or yung equivalent ng MTAP. And I also do review program for certification examination takers. So, ang ano kasi dun, merong sariling TOS ang certification examination for NMT or Nuclear medicine technology. Five cluster siya… yun yung composition ng five clusters. So, I think that's cluster 1, pwede mong isulat yan (Interviewer: okay po sir)… hindi ko alam anong extent mo siya ipapasok sa discussion eh. Pero kunwari, ang cluster 1 is radiation physics, number 2 ay radiation protection, number 3 is instrumentation and quality…QA, QC.. number 4, radiopharmaceuticals and pharmaceuticals. And then 5 yung clinical procedures. Yun yung limang clusters. So, kung tatanungin mo ako kung paano siya i-integrate sa curriculum, uhm, magdagdag ng limang subjects, atleast yung limang subjects because they are the core competencies na needed sa profession.

Interviewer (Aborje)

- opo
- So, nabanggit niyo po na parang mag another course such as lec and lab, tas lima pong sections.

Sir Alfred Bayan (RMT-10-260313-1727)

- Oo
- Hindi ko na sasagutin yung ano, ilang unit ba per course (Interviewer: opo)
- I think out of scope na siya dito sa discussion natin
- Kasi very into curriculum yung sagot ng mga yan.
- Pero kung idagdag ko naman, for context. Ngayon, I'm on my 3rd year as a full-time professor para atleast ano… so, kung isasama ko siya as my nuclear medicine practice, 10 years na ako (Interviewer: ah yes po, kasi 7 tapos 3). 7 siya directly sa nucmed facility pero 3 years as academe.

Interviewer (Aborje)

- Yes po
- If it's okay to ask po sir, saan po kayo nagtuturo?

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah… sa ano, sa De La Salle Medical and health sciences institute or DLSMHSI sa cavite. (interviewer: ooohhh, that's nice po)… you can search that hehe…it is still the only institute offering the BSNMT program.

Interviewer (Aborje)

- Opo
- So, to wrap up… papunta na po tayo sa wrap-up questions sir, actually, ang bilis po natin
- So, nasagot niyo na po ito kanina, pero….. So, overall, would you recommend the formal incorporation of nuclear medicine topics in the medtech curriculum po?

Sir Alfred Bayan (RMT-10-260313-1727)

- Ah… overall… would I recommend? Uhm, honestly, not totally …agree. Hindi ko alam kung paano niyo siya i-rerelate noh? Kasi, kung i-bibased on my current practices sa academe, ayun, suggest otherwise to create a separate undergraduate program such as BSNMT. Pero if you would really, incorp[orate that in the MT practice, medyo madaming kailanagn gawin doon. When it comes to MT curriculum, magdagdag atleast ng limang binanggit ko.

Interviewer (Aborje)

- Opo
- So, is there anything po, you want to share your experience that would help us understand the alignment between the Medtech curriculum and nuclear medicine practice?

Sir Alfred Bayan (RMT-10-260313-1727)

- Oo, ang ano ko lang naman, ito, personal ano ko lang naman, opinion aside sa mga sinabi ko. Maganda na ang background ko is medtech, kasi kung… sa totoo lang, ang nuclear medicine procedures naman ay mostly heavy on physiology.. so, more than the anatomy… more on physiology. And very heavy tayo sa concepts na yun.
- Uhm chemistry,.. sa part palang ng chemistry and biochemistry, very heavy na ang range natin dun. So, oo.. maiintindihan natin yun kung ano yung nangyayari doon. Kaya binanggit ko, in-emphasize ko yung endocrinology kasi that's one topic when it comes to physiology.. noh.. or body functions . So very important. Kasi ang nuclear medicine procedures naman ay … into the physiology of the body… pati physiology ng disorders… ah diseases rather…for example, sa oncology… mga yun.

- So, very ano, appropriate siya as a foundation for nuclear medicine technology profession or practice. Kaya lang, of course sa technical side, madami tayomng kailangan i-catch up… pagmamanipulate palng ng mga machine, hindi ko na alam first day of nuclear medicine technology… yun…

- Paano pa yung pag-process, mga ganon.. matagal, matagal ko siynag ano.. ako pa naman yung ano yung medyo may pagka-OC… kaya sinabi ko 2 years bago ako naging confident… kasi gusto ko, perfect ko siya na pwede ng walang magbantay sa akin… bago ko masabi na okay na ako. Yun

Interviewer (Aborje)

- That's nice po sir. So lastly po, if may marerecommend ka for our reseach, who else do you think we should speak to? In order to understand this topic better po.

Sir Alfred Bayan (RMT-10-260313-1727)

- Oo, uhm, kung same professions, specificly ba? Kasi MT lang eh… Meron naman ako marerecommend, uhm… kung currently…adviser na ata… si ma'am angie… tam aba adviser niyo si ma'am angie? (interviewer: anong surname po sir?) … angeline… sory hindi ko kabisado apelyido ni ma'am… (interviewer: ah, ma'am Rillorta po?)… oo… tama… tama.. tama… So kung ano… siya ba ang adviser niyo? Hindi? (interviewer: yeah.. actually siya po yung ano, yung research FRIP po naming)… oo.. So, kung hindi siya, ahh… si ma'am pat ochoa…ma'am Patricia Ochoa kung hindi siya nasa BGH, andiyan pa siya sa sacred heart as nuclear medicine technologist. Pero kung previously working, meron akong friend… close friend… kaso doctor na siya eh.. so kung pasok siya sa criteria niyo, I think pwede ko siya i-recommend.. yung ka-batch ko si, Jenon Mangawa. (interviewer: actually sir, nabanggit ka ni sir, kasi na-interview na po naming siya kanina)… ah okay…hehehe (interviewer: nabanggit niyo po na best friends daw po kayo)… oo… oo… nainterview na palay un di ako sinabihan… ah so, ilang years din yung nag-work.. teka… pwede ako mag recommend.. hindi natin alumni… pero of course ipaalam ko muna … kung anoo.. previous ka-trabaho ko siya from St. Lukes QC… 9Interviewer: ooohh, hopefuilly po pasok siya sa criteria po ng research namin) … currently working parin siya… I think more than 5 years na siyang nag wowork don.. oo… mga 7 years na din siguro siya

- Tapos, pwede din supervisor… kaso puro St. Lukes yung marerecommend ko.

Interviewer (Aborje)

- Opo

- Okay sir.. So, actually tapos na po yung mga questions po naming.

- Before ko naming i-conclude po tong interview na to sir, do you have any questions po for us ? Or is there anything we would like to add?

Sir Alfred Bayan (RMT-10-260313-1727)

- Wala na.. okay na ko.. okay na ko..
- Okay, so with that po, I'd like to thank you once again for taking the time to share your insights on how medical technology professionals perceive the integration of nuclear medicine topics in the medtech curriculum po. So, your responses have been incredibly helpful to our study. This ends our interview, which took about maybe less than 20 minutes po, so we'll now stop the recording po.

Noeimi Fermato:

Background:

My last role there was a safety protection officer.

Follow up:
The last position was a safety protection officer, right?

- RPO, radiation protection officer.

How long have you been working in this field?

- Five years as an NMT and as an RPO recently. Two years as an NMT for four years and then the rest is RPO. So both of those roles in total are five years, Two years as a regular nuclear medicine technologist. After two years, assistant radiation protection officer for almost two years and then remaining RPO, radiation protection officer.

I. During your undergraduate education under the MEDTEP curriculum, to what extent were you exposed to the nuclear medicine concepts like radiopharmaceutical, radiation safety, and etc.?

- Malapit sa MEDTEP curriculum which is yung real laboratory. ara siyang CC, clinical chemistry but very specialized and still using radiopharmaceuticals. So yung sa atin pag CC, clinical chemistry meron meron yung routine chemistry atsaka special chemistry. So sa nuclear medicine kasi meron silang special chem which is yun yung mostly sa thyroid function. So the tests are FT4, TSH, FT3, thyroglobulin. Pero ang pinagkaiba kasi nila sa MEDTEP sa atin, sa NUC MED gumagamit ng radiation. So may radiation yung mga reagents nila. So I think that's the only thing na medyo malapit between MEDTEP atsaka nuclear medicine. And the rest is medyo malayo actually.

Ii. In your opinion, does the current MEDTEP curriculum or the curriculum you were in, does it prepare graduates well for working in the nuclear medicine including instrumentation or work processes?

- think 50-50. Kasi very kwan naman yung curriculum natin, yung sa MEDTEP curriculum. It prepares you mentally atsaka work-wise. Parang ready ka to take it. Pero with the kwan knowledge, medyo kailangan pa siyang ma-widen maliban lang sa MEDTEP.Kasi maraming term and other kwan na hindi natin napag-aralan nung nasa MEDTEP curriculum tayo. So I think hindi siya enough na you graduate as MEDTEP then tapon agad sa nuclear medicine. I think need dalaga ng kwan. Need dagdagan or supplemented. Parang subjects na kailangan idagdag para mas ma-prepare yung MEDTEP kung gusto niyang maging part ng nuclear medicine.

Follow up:

As you began working, what knowledge aspects or skills felt missing from the MEDTEP curriculum?

- So when I started working as a nuclear medicine technologist, madaming terms na hindi familiar. And then the machines na hindi natin na gamit or na college, hindi natin siya nakita or nagalaw. So more on kwan siguro sa machineries at saka sa mga terms sa nucemed. Kasi more on radial sila.

Did these gaps that you have mentioned in knowledge or skills, did they affect your early performance or confidence in the field?

- Yes. Actually medyo kwan din ako nung first law sa nucmed. Nahirapan akong gumalaw kasi takot nga akong gumalaw mag-isa or unsupervised since hindi familiar yung mga machines and yung mga work. Parang nung college tayo or during internship, kwan tayo eh, naisabak tayo dun sa mga gagawin pag nag-work ka na. So nung nag-start ako sa nucmed, hindi siya familiar. So yun, medyo alanganin ako nung umpisa gumalaw kasi hindi ko pa siya alam. Pero yun nga. Meron naman kwan, meron namang course na ikot na kwan ang nucmed para mas malawak pa yung kwan mo. Para matawag kang nuclear medicine technologist, kailangan mong mag-undergo actually ng course.

Iii From your perspective, what minimum knowledge or skills from the nuclear medicine field should medtech graduates have before entering this field?

- Minimum na skills. Actually, yung siguro pinaka-basic is the phlebotomy skills. Kasi for nuclear medicine, mga patients is usually mga cancer patient or mga patient with thyroid problems and it requires us to extract blood and also to administer blood actually. So during medtech natin, we only draw blood. Pero during nuclear medicine, kailangan mo ring mag-inject. So administer ng dugo. So yun yung pinaka-basic na kailangan. Siguro yun yung phlebotomy skills.

Iv. With regards to the whole topic under the medtech curriculum, are there topics that you find most valuable in your current work in the nuclear medicine field? And if yes po, ano pong specific na topics?

- Siguro, Kwan, thyroid function. Kasi most ng Kwan natin, meron kasi sa nuclear medicine may tinatawag na rea-lab. Para siyang yung clinical chemistry lab. So yun. So routine work there is to extract blood and then run the test. Parang sa CC lang din. So ayun. During running the test, kailangan familiar ka sa mga blood results. So yun. Alam mo dapat yung result. Parang alam mo dapat na Kwan siya. Ano pong term dito? Interpret po. Oo, interpret. Kasi may mga doctors. Marami kasing doctors sa nucmed. Na ayun. Medyo Kwan sila sa results. So kailangan. Kunwari yung FT4 atsaka TSH. So yun yung mga topics na under MT Curriculum na valuable during nuclear medicine field.

V. During your transition into the nuclear medicine field, what competencies in the nuclear medicine field did you find difficult or learn to adjust to?

- Yung mixing ng mga radiopharmaceuticals, administering and also scanning. Kasi yung mga machine nila sa scan very different or hindi talaga natin na touch nung medtech tayo. So yun. So sa nucmed meron silang tinatawag na hot lab. So dun pinagprepare yung mga radio pharmaceuticals tayo na administer sa patient. So nung una medyo nahihirapan ako kasi kailangan niyo siyang standard. And then very careful kasi handling of regeneration na siya. So yun yung mahirap pag-adjustan and yan. So kapag i-administer din sa patient, syempre hindi natin na gawa yun nung medtech tayo. Since drawing of blood lang usually. So medyo adjustment din yan kasi hindi ako sure kung ano yung mga effect or ano yung mga possibilities kapag mali yung administration ng gamot. And then sa scanning naman, siguro yung interpretation kasi we were just scanning and then hindi namin alam kung ano scan namin kasi syempre hindi natin na touch yun during our medtech.